

Can Heritage Texts Reveal Their Category?

Machine Learning Task for World Heritage Classification

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An aerial photograph of a two-lane asphalt road that curves through a dry, hilly landscape. The hills are covered in sparse, low-lying vegetation and small shrubs. The sky is overcast and grey. The overall tone of the image is muted and atmospheric.

01

Project Overview

Research Question

Original Question: Can the Outstanding Universal Value (OUV) criteria predict the World Heritage category, and vice versa, and does justification text improve this prediction?

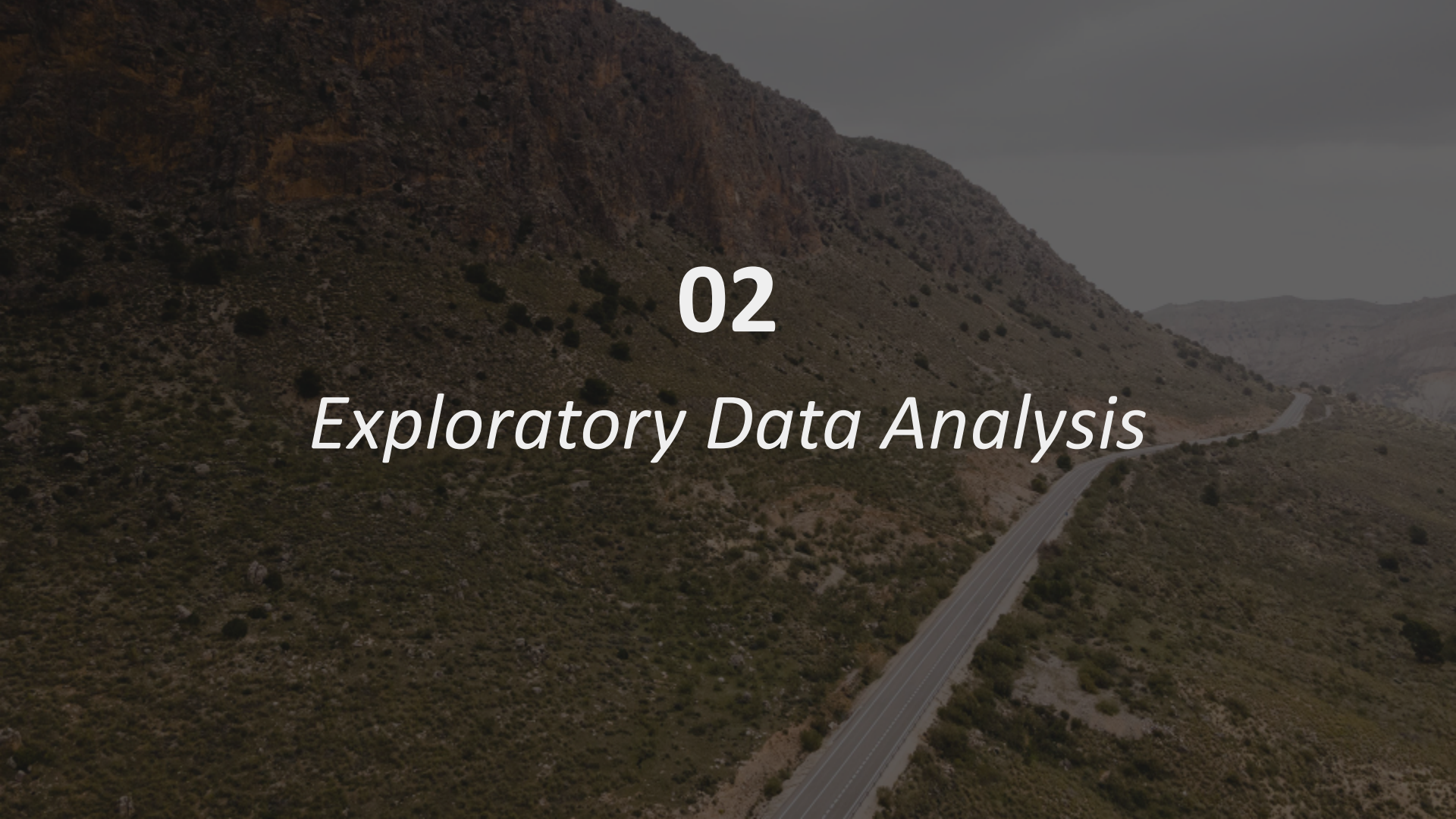
Revised Question: Can we use the World Heritage description and justification texts to predict their category (Cultural/Natural)?

Machine Learning Task

Classification Task: To train a machine learning model to classify World Heritage Sites into their corresponding Heritage Category (Cultural/Natural) using description and justification text.

Objective and Goals

The main objective is to develop a machine learning model that accurately classifies world heritage sites through texts, and to explore whether machine learning can identify patterns in how cultural and natural heritage values are described, which may help the public understand how heritage institutions structure and communicate heritage values.

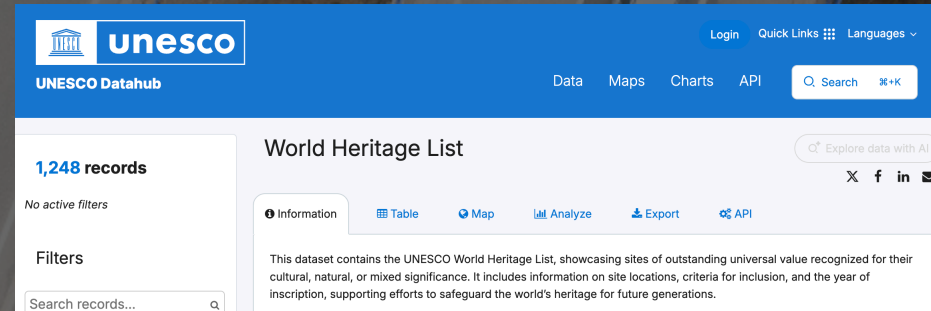
An aerial photograph of a two-lane asphalt road that curves through a dry, hilly landscape. The hills are covered in sparse, low-lying vegetation and small shrubs. The sky is overcast and grey. The overall tone of the image is muted and atmospheric.

02

Exploratory Data Analysis

Data Collection

- Data source: UNESCO World Heritage List
- Data size: 1,248 samples
- Data format: mixed data type
 - text: {Names, Description, Justification ...}
 - categorical: {Criteria, Category, States Names, Region...}
 - numerical: {Date inscribed, ID, Coordinates, Components Count ...}
- Possible features that could represent the data
 - OUV criteria data, indicated by the Criteria column in the dataset; each criterion is binary, but each combination is multi-labeled.
 - World Heritage Category, indicated by Category/Category ID.
 - Justification Text, indicated by the Justification column in the data; in addition, the Description column can also be utilized for more information.



Data Collection

- Data source: UNESCO World Heritage List

	Name EN	Name FR	Name ES	Name RU	Name AR	Name ZH	Short Description EN	Short Description FR	Short Description ES	Short Description RU	...	Region	Region Code
0	The works of Jože Plečnik in Ljubljana – Human...	Les œuvres de Jože Plečnik à Ljubljana – une c...	Obras de Jože Plečnik en Liubliana – una conce...	Работы Йоже Плечника в Любляне – городской диэ...	أعمال جوزيه بليشكن في مدينة ليوبليانا – تصميم ...	卢布爾雅那的约热·普利奇尼克作品——以人为本的城市设计	The works of Jože Plečnik carried in Ljubljana be...	Les œuvres de Jože Plečnik, réalisées à Ljublj...	La obra que Jože Plečnik llevó a cabo en Liubl...	Работы Йоже Плечника, выполненные в Любляне в ...	...	Europe and North America	EUR
1	Fossil Hominid Sites of South Africa	Sites des hominidés fossiles d'Afrique du Sud	Sitios de homínidos fósiles de Sudáfrica	Стеркфонтейн, Сварткранс, Кромдрай и окрестнос...	موقع الهومينيدات (متحجرات الإنسان البدائي) في ...	斯泰克方丹、斯瓦特兰斯、科罗姆德拉伊和维罗恩斯的化石遗址	The Taung Skull Fossil Site, part of the exten...	C'est sur ce site que le célèbre crâne fossile...	Fue en este sitio donde se encontró, en 1924, ...	Таунг-Скал-Фоссил-Сайт – это одно из добавлени...	...	Africa	AFR
2	Robben Island	Robben Island	Robben Island	Остров Роббен-Айленд	جزيرة روبن	罗布恩岛	Robben Island was used at various times between...	Robben Island a été utilisée à différentes épo...	Robben Island fue utilizada en diferentes époc...	В период XVII-XX вв. Роббен-Айленд использовал...	...	Africa	AFR
3	The Emergence of Modern Human Behaviour: The P...	L'émergence du comportement humain moderne : l...	La aparición del comportamiento humano moderno...	Возникновение поведения людей современного тип...	ظهور السلوك البشري الحديث: مستوطنات العصر الحدد	现代人的出现：南非更新世人类遗址	This serial property contributes to the unders...	Ce bien en série contribue à la compréhension ...	Este sitio en serie contribuye a la comprensió...	Этот серийный объект позволяет составить предс...	...	Africa	AFR
4	#Khomani Cultural Landscape	Paysage culturel des #Khomani	Paisaje cultural de los #khomani	NaN	منظر الخوماني الثقافي	NaN	The #Khomani Cultural Landscape is located at ...	Le paysage culturel des #Khomani est situé à l...	Este paisaje cultural se sitúa en la frontera ...	NaN	...	Africa	AFR

...}

et; each criterion is binary, but

ta; in addition, the Description

column can also be utilized for more information.

Data Collection

- Data source: UNESCO World Heritage List

	Name EN	Name FR	Name ES	Name RU	Name AR	Name ZH	Short Description EN	Short Description FR	Short Description ES	Short Description RU	...	Region	Region Code						
0	The works of Jože Plečnik in Ljubljana – Human...	Les œuvres de Jože Plečnik à Ljubljana – une c...	Obras de Jože Plečnik en Ljubljana – una conce...	Работы Йоже Плечника в Любляне – городской диза...	أعمال جوزيه بليشكن في مدينة ليوبليانا – تصميم ...	卢布尔雅那的约热·普利奇尼克作品——以人为本的城市设计	The works of Jože Plečnik	Les œuvres de Jože Plečnik	La obra que Jože Plečnik	Работы Йоже Плечника, выполненные ...	...	Europe and North...	EUR						
							Transboundary	Main Image					Images		UUID	ID	Coordinates	Components	Components Count
1	Fossil Hominid Sites of South Africa	Sites des hominidés fossiles d'Afrique du Sud	Sitios de homínidos fósiles de Sudáfrica	Стеркфонтейн, Сварткрэнс, Кромдрай и окрестнос...	موقع الهومينيدات (متحجرات) الإنسان البدائي في ...	斯泰克方丹、斯瓦特兰斯、科罗姆德拉伊和维罗恩斯的化石遗址	False	https://data.unesco.org/api/explore/v2.1/catalog/record/181314	https://whc.unesco.org/document/181314/site_16...		0faea55c-17b3-5c43-9ccb-784ef7071aa6	1643	46.0433333333, 14.5022222222	{name: Promenade along the Embankments and Bri...		7			
2	Robben Island	Robben Island	Robben Island	Остров Роббен-Айленд	جزيرة روبن	罗布恩岛	False	https://data.unesco.org/api/explore/v2.1/catalog/record/113350	https://whc.unesco.org/document/113350/site_09...		e381540f-9349-54cf-8998-ef85cc4a5d51	915	-24.15861, 29.17694	{name: Makapan Valley, ref: 915bis-002, latitu...		3			
3	The Emergence of Modern Human Behaviour: The P...	L'émergence du comportement humain moderne : l...	La aparición del comportamiento humano moderno...	Возникновение поведения людей современного типа...	ظهور السلوك البشري الحديث: مستوطنات العصر الحدد...	现代人的出现：南非更新世人类遗址	False	https://data.unesco.org/api/explore/v2.1/catalog/record/113362	https://whc.unesco.org/document/113362/site_09...		3c4fd4bf-395c-522b-b349-858d91573e3e	916	-33.8, 18.36666667	{name: Robben Island, ref: 916, latitude: -33....		1			
4	#Khomani Cultural Landscape	Paysage culturel des #Khomani	Paisaje cultural de los #khomani	NaN	منظر الخوماني الثقافي	NaN	False	https://data.unesco.org/api/explore/v2.1/catalog/record/198607	https://whc.unesco.org/document/198607/site_17...		afa30144-9e1f-51c0-bb7c-183041c5ddcb	1723	-32.3863888889, 18.4525	{name: Sibhudu Cave, ref: 1723-003, latitude: ...		3			
							False	https://data.unesco.org/api/explore/v2.1/catalog/record/158113	https://whc.unesco.org/document/158113/site_15...		dad9f15c-9151-522b-a4e6-894b5020c161	1545	-25.6876111111, 20.3745833333	{name: #Khomani Cultural Landscape, ref: 1545,...		1			

column can also be utilize

Data Cleaning

- Data source: UNESCO World Heritage List
- Data size: 1,248 samples
- Data format: mixed data type

	Name EN	Short Description EN	Justification EN	Criteria	Category
0	The works of Jože Plečnik in Ljubljana – Human...	The works Jože Plečnik carried in Ljubljana be...	Brief synthesis The urban design for Ljubljana...	(iv)	Cultural
1	Fossil Hominid Sites of South Africa	The Taung Skull Fossil Site, part of the exten...	Brief synthesis The undulating landscape conta...	(iii)	Cultural
2	Robben Island	Robben Island was used at various times betwee...	Brief synthesis Robben Island was used at vari...	(iii)	Cultural
3	The Emergence of Modern Human Behaviour: The P...	This serial property contributes to the unders...	Brief synthesis Diepkloof Rock Shelter, Pinnac...	(iii)(iv)(v)	Cultural
4	ǀKhomani Cultural Landscape	The ǀKhomani Cultural Landscape is located at ...	Brief synthesis The ǀKhomani Cultural Landscap...	(v)	Cultural
5	Catalan Romanesque Churches of the Vall de Boí	The narrow Vall de Boí is situated in the high...	Brief synthesis The Vall de Boí is located in ...	(ii)(iv)	Cultural
6	Cave of Altamira and Paleolithic Cave Art of N...	Seventeen decorated caves of the Paleolithic a...	Brief synthesis The caves of Altamira, Peña de...	(i)(iii)	Cultural
7	Historic Walled Town of Cuenca	Built by the Moors in a defensive position at ...	Brief synthesis Cuenca is a municipality locat...	(ii)(v)	Cultural
8	Historic City of Toledo	Successively a Roman municipium, the capital o...	Brief synthesis Toledo is a city located in ce...	(i)(ii)(iii)(iv)	Cultural
9	Garajonay National Park	Laurel forest covers some 70% of this park, si...	Brief synthesis Not far off the north-west coa...	(vii)(ix)	Natural

- World Heritage Category, indicated by Category/Category ID.
- Justification Text, indicated by the Justification column in the data; in addition, the Description column can also be utilized for more information.

Data Cleaning

- Data source: UNESCO World Heritage List
- Data size: 1,24
- Data format: n

```

null_sum = df.isnull().sum()
null_sum
✓ 0.0s

```

```

Name EN      0
Short Description EN  0
Justification EN  50
Criteria      16
Category      0
dtype: int64

```

Criteria	Category
----------	----------

0	The works of Jože Plečnik in Lj	(iv)	Cultural
1	Fossil Hominid Si	(iii)	Cultural
2		(iii)	Cultural
3	The Emergence of Modern Human	(iii)(iv)(v)	Cultural
4	‡Khomani (We can see that although some samples have no Justification text, all sites have at least Short Description, which is viable for text classification.	(v)	Cultural
5	Catalan Romanesque Churches	(ii)(iv)	Cultural
6	Cave of Altamira and Paleolithic	(i)(iii)	Cultural
7	Historic Walk	(ii)(v)	Cultural
8	Hist	(i)(ii)(iii)(iv)	Cultural
9	Garaj	(vii)(ix)	Natural

```

df['Justification EN'] = df['Justification EN'].fillna('')
✓ 0.0s

```

```

# Drop samples that have no "Criteria" values, either NaN or blank
df = df.dropna(subset=["Criteria"])
df = df[df["Criteria"].str.strip() != ""]
df = df.reset_index(drop=True)
df.shape
✓ 0.0s

```

```

(1232, 5)

```

Description

- World He
- Justificati

column can also be utilized for more information.

Data Processing

EDA

Categorical Data

	Criteria	Category
0	(iv)	Cultural
1	(iii)	Cultural
2	(iii)	Cultural
3	(iii)(iv)(v)	Cultural
4	(v)	Cultural
...	...	...
1227	(i)(iii)	Cultural
1228	(iv)(v)	Cultural
1229	(iii)(iv)	Cultural
1230	(vii)(ix)(x)	Natural
1231	(iii)(v)	Cultural

1232 rows × 2 columns

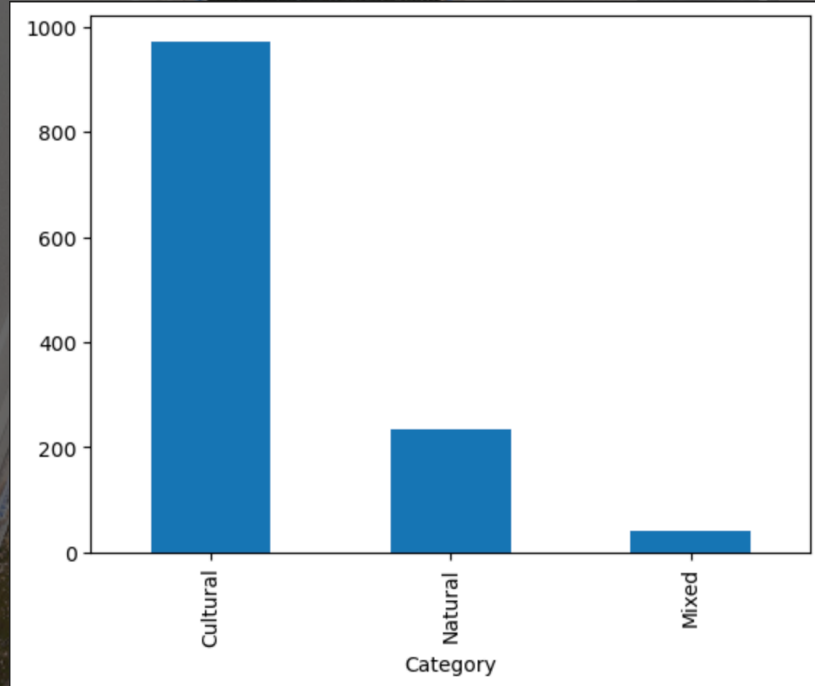
Data Processing

Categorical Data

	Criteria	Category
0	(iv)	Cultural
1	(iii)	Cultural
2	(iii)	Cultural
3	(iii)(iv)(v)	Cultural
4	(v)	Cultural
...	...	...
1227	(i)(iii)	Cultural
1228	(iv)(v)	Cultural
1229	(iii)(iv)	Cultural
1230	(vii)(ix)(x)	Natural
1231	(iii)(v)	Cultural

1232 rows x 2 columns

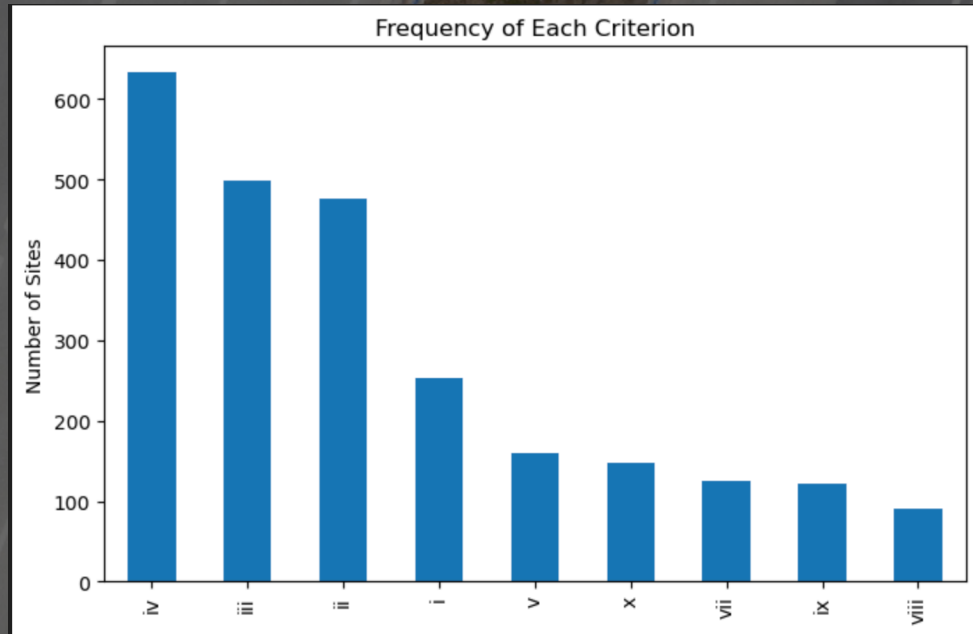
```
Category
Cultural    956
Natural     235
Mixed        41
Name: count, dtype: int64
```



Data Processing

Categorical Data

	Criteria	Category
0	(iv)	Cultural
1	(iii)	Cultural
2	(iii)	Cultural
3	(iii) (iv) (v)	Cultural
4	(v)	Cultural
...	...	...
1007	(iv) (v)	Cultural



	Name EN	Short Description EN	Justification EN	Category	i	ii	iii	iv	ix	v	vii	viii	x
0	The works of Jože Plečnik in Ljubljana – Human...	The works Jože Plečnik carried in Ljubljana be...	Brief synthesis The urban design for Ljubljana...	Cultural	0	0	0	1	0	0	0	0	0
1	Fossil Hominid Sites of South Africa	The Taung Skull Fossil Site, part of the exten...	Brief synthesis The undulating landscape conta...	Cultural	0	0	1	0	0	0	0	0	0
2	Robben Island	Robben Island was used at various times betwee...	Brief synthesis Robben Island was used at vari...	Cultural	0	0	1	0	0	0	0	0	0
3	The Emergence of Modern Human Behaviour: The P...	This serial property contributes to the unders...	Brief synthesis Diepkloof Rock Shelter, Pinnac...	Cultural	0	0	1	1	0	1	0	0	0
4	#Khomani Cultural Landscape	The #Khomani Cultural Landscape is located at ...	Brief synthesis The #Khomani Cultural Landscap...	Cultural	0	0	0	0	0	1	0	0	0
5	Catalan Romanesque Churches of the Vall de Boí	The narrow Vall de Boí is situated in the high...	Brief synthesis The Vall de Boí is located in ...	Cultural	0	1	0	1	0	0	0	0	0
6	Cave of Altamira and Paleolithic Cave Art of N...	Seventeen decorated caves of the Paleolithic a...	Brief synthesis The caves of Altamira, Peña de...	Cultural	1	0	1	0	0	0	0	0	0
7	Historic Walled Town of Cuenca	Built by the Moors in a defensive position at ...	Brief synthesis Cuenca is a municipality locat...	Cultural	0	1	0	0	0	1	0	0	0
8	Historic City of Toledo	Successively a Roman municipium, the capital o...	Brief synthesis Toledo is a city located in ce...	Cultural	1	1	1	1	0	0	0	0	0
9	Garajonay National Park	Laurel forest covers some 70% of this park, si...	Brief synthesis Not far off the north-west coa...	Natural	0	0	0	0	1	0	1	0	0

Data Processing

Categorical Data: Heritage Category as Target Variable

Binarize the target variable

```
df['label'] = df['Category'].map({'Cultural':1, 'Natural':0})
display(df.head(10))
```

✓ 0.0s

Python

	Name EN	Short Description EN	Justification EN	Category	label
0	The works of Jože Plečnik in Ljubljana – Human...	The works Jože Plečnik carried in Ljubljana be...	Brief synthesis The urban design for Ljubljana...	Cultural	1
1	Fossil Hominid Sites of South Africa	The Taung Skull Fossil Site, part of the exten...	Brief synthesis The undulating landscape conta...	Cultural	1
2	Robben Island	Robben Island was used at various times betwee...	Brief synthesis Robben Island was used at vari...	Cultural	1
3	The Emergence of Modern Human Behaviour: The P...	This serial property contributes to the unders...	Brief synthesis Diepkloof Rock Shelter, Pinnac...	Cultural	1
4	ǀKhomani Cultural Landscape	The ǀKhomani Cultural Landscape is located at ...	Brief synthesis The ǀKhomani Cultural Landscap...	Cultural	1
5	Catalan Romanesque Churches of the Vall de Boí	The narrow Vall de Boí is situated in the high...	Brief synthesis The Vall de Boí is located in ...	Cultural	1
6	Cave of Altamira and Paleolithic Cave Art of N...	Seventeen decorated caves of the Paleolithic a...	Brief synthesis The caves of Altamira, Peña de...	Cultural	1
7	Historic Walled Town of Cuenca	Built by the Moors in a defensive position at ...	Brief synthesis Cuenca is a municipality locat...	Cultural	1
8	Historic City of Toledo	Successively a Roman municipium, the capital o...	Brief synthesis Toledo is a city located in ce...	Cultural	1
9	Garajonay National Park	Laurel forest covers some 70% of this park, si...	Brief synthesis Not far off the north-west coa...	Natural	0

Data Processing

Text Data: Justification and Description as Feature Variables

```
df['description'] = clean_text(df, "Short Description EN")
```

```
df.head()
```

✓ 0.0s

Python

100% | 1191/1191 [00:00<00:00, 27881.65it/s]

	Name EN	Short Description EN	Justification EN	Category	label	length_d	length_j	description
0	The works of Jože Plečnik in Ljubljana – Human...	The works Jože Plečnik carried in Ljubljana be...	Brief synthesis The urban design for Ljubljana...	Cultural	1	1295	8102	The works Jože Plečnik carried in Ljubljana be...
1	Fossil Hominid Sites of South Africa	The Taung Skull Fossil Site, part of the exten...	Brief synthesis The undulating landscape conta...	Cultural	1	710	6093	The Taung Skull Fossil Site part of the extens...
2	Robben Island	Robben Island was used at various times between...	Brief synthesis Robben Island was used at vari...	Cultural	1	346	4646	Robben Island was used at various times between...
3	The Emergence of Modern Human Behaviour: The P...	This serial property contributes to the unders...	Brief synthesis Diepkloof Rock Shelter, Pinnac...	Cultural	1	766	5489	This serial property contributes to the unders...
4	#Khomani Cultural Landscape	The #Khomani Cultural Landscape is located at ...	Brief synthesis The #Khomani Cultural Landscap...	Cultural	1	716	10358	The #Khomani Cultural Landscape is located at ...

```
df["justification"] = clean_text(df, "Justification EN")
```

```
df.head()
```

✓ 1.1s

Python

100% | 1191/1191 [00:01<00:00, 1088.98it/s]

	Name EN	Short Description EN	Justification EN	Category	label	length_d	length_j	description	justification
0	The works of Jože Plečnik in Ljubljana – Human...	The works Jože Plečnik carried in Ljubljana be...	Brief synthesis The urban design for Ljubljana...	Cultural	1	1295	8102	The works Jože Plečnik carried in Ljubljana be...	Brief synthesis The urban design for Ljubljana...
1	Fossil Hominid Sites of South Africa	The Taung Skull Fossil Site, part of the exten...	Brief synthesis The undulating landscape conta...	Cultural	1	710	6093	The Taung Skull Fossil Site part of the extens...	Brief synthesis The undulating landscape conta...
2	Robben Island	Robben Island was used at various times between...	Brief synthesis Robben Island was used at vari...	Cultural	1	346	4646	Robben Island was used at various times between...	Brief synthesis Robben Island was used at vari...
3	The Emergence of Modern Human Behaviour: The P...	This serial property contributes to the unders...	Brief synthesis Diepkloof Rock Shelter, Pinnac...	Cultural	1	766	5489	This serial property contributes to the unders...	Brief synthesis Diepkloof Rock Shelter Pinnac...
4	#Khomani Cultural Landscape	The #Khomani Cultural Landscape is located at ...	Brief synthesis The #Khomani Cultural Landscap...	Cultural	1	716	10358	The #Khomani Cultural Landscape is located at ...	Brief synthesis The #Khomani Cultural Landscap...

Data Processing

Text Data: Justification and Description as Feature Variables

```
df['description'] = clean_text(df, "Short Description EN")
```

```
from sklearn.feature_extraction.text import TfidfVectorizer

# Vectorizer for Description
vectorizer_desc = TfidfVectorizer(stop_words=stopwords, max_df=0.9, min_df=3, ngram_range=(1,1))
X_desc = vectorizer_desc.fit_transform(df['description'])

# Vectorizer for Justification
vectorizer_just = TfidfVectorizer(stop_words=stopwords, max_df=0.9, min_df=3, ngram_range=(1,1))
X_just = vectorizer_just.fit_transform(df['justification'])
```

✓ 0.6s

Python

```
feature_names_desc = vectorizer_desc.get_feature_names_out()
feature_names_just = vectorizer_just.get_feature_names_out()
combined_feature_names = list(feature_names_desc) + list(feature_names_just)
```

✓ 0.0s

Python

```
print("X_desc.shape : ",X_desc.shape)
print("X_just.shape : ",X_just.shape)
```

✓ 0.0s

Python

```
X_desc.shape : (1191, 3224)
X_just.shape : (1191, 11623)
```


Finalizing Feature Matrix

- Construct the matrix from the processed textual features and the categorical target variable
- Split the data matrix into training and testing with a 70:30 ratio to prepare for model training
- Shuffle the data at the same time (random permutations of the collections)
- Apply *RandomUnderSampler* only on the training data to make it balanced and remain test data unchanged to evaluate performance on a realistic distribution
- Assign a random state of 42 for reproducible output

```
from scipy.sparse import hstack
```

```
X = hstack([X_desc, X_just])
```

✓ 0.0s

Python

Finalizing Feature Matrix

- Construct the matrix from the processed textual features and the categorical target variable
- Split the data matrix into training and testing with a 70:30 ratio to prepare for model training
- Shuffle the data at the same time (random permutations of the collections)
- Apply *RandomUnderSampler* only on the training data to make it balanced and remain test data unchanged to evaluate performance on a realistic distribution
- Assign a random state of 42 for reproducible output

```
from scipy.sparse import hstack
```

```
X = hstack([X_desc, X_just])
```

✓ 0.0s

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
                                                    shuffle=True,
                                                    random_state=42)

# check data size after splitting
print("Training data: X : {}, y : {}".format(X_train.shape, y_train.shape))
print("Testing data: X : {}, y : {}".format(X_test.shape, y_test.shape))
```

✓ 0.0s

Python

Training data: X : (833, 14847), y : (833,)

Testing data: X : (358, 14847), y : (358,)

Finalizing Feature Matrix

- Construct the matrix from the processed textual features and the categorical target variable
- Split the data matrix into training and testing with a 70:30 ratio to prepare for model training
- Shuffle the data at the same time (random permutations of the collections)
- Apply *RandomUnderSampler* only on the training data to make it balanced and remain test data unchanged to evaluate performance on a realistic distribution
- Assign a random state of 42 for reproducible output

```
from scipy.sparse
```

```
X = hstack([X_c
```

✓ 0.0s

```
from imblearn.under_sampling import RandomUnderSampler
```

```
rus = RandomUnderSampler(random_state=42)
```

```
X_train_res, y_train_res = rus.fit_resample(X_train, y_train)
```

```
print(sorted(Counter(y_train_res).items()))
```

✓ 0.5s

Python

```
test_split
```

```
est_split(X, y, test_size=0.3,
```

```
shuffle=True,
```

```
random_state=42)
```

```
at(X_train.shape, y_train.shape))
```

```
t(X_test.shape, y_test.shape))
```

✓ 0.0s

Python

Training data: X : (833, 14847), y : (833,)

Testing data: X : (358, 14847), y : (358,)

An aerial photograph of a two-lane asphalt road that curves through a dry, hilly landscape. The hills are covered in sparse, low-lying vegetation and small shrubs. The sky is overcast and grey. The overall tone of the image is muted and atmospheric.

03

Model Exploration



Random Forest

```
from sklearn.model_selection import GridSearchCV

rf = RandomForestClassifier(random_state=42)
hy_parameters = {'criterion':['entropy', 'gini'],
                 'n_estimators': [10, 100, 200, 1000]} # 2*4 = 8 combinations

grid_cv = GridSearchCV(estimator = rf, param_grid = hy_parameters, cv=5)

grid_cv.fit(X_train_res, y_train_res)
```

✓ 12.9s

Python

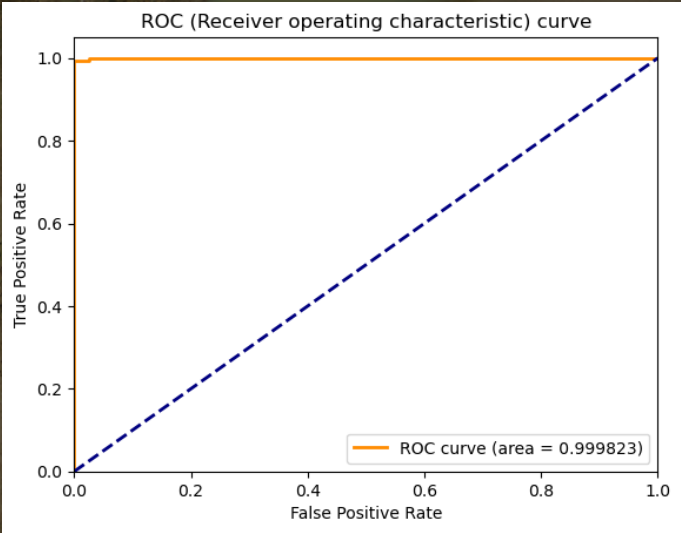
Random Forest

- number of trees (*n_estimators*)
- splitting criterion (*gini* vs *entropy*)
- Using around **1000** trees improved model stability
- ***entropy*** performed better than *gini*

	precision	recall	f1-score	support
Natural	0.96	1.00	0.98	82
Cultural	1.00	0.99	0.99	276
accuracy			0.99	358
macro avg	0.98	0.99	0.99	358
weighted avg	0.99	0.99	0.99	358

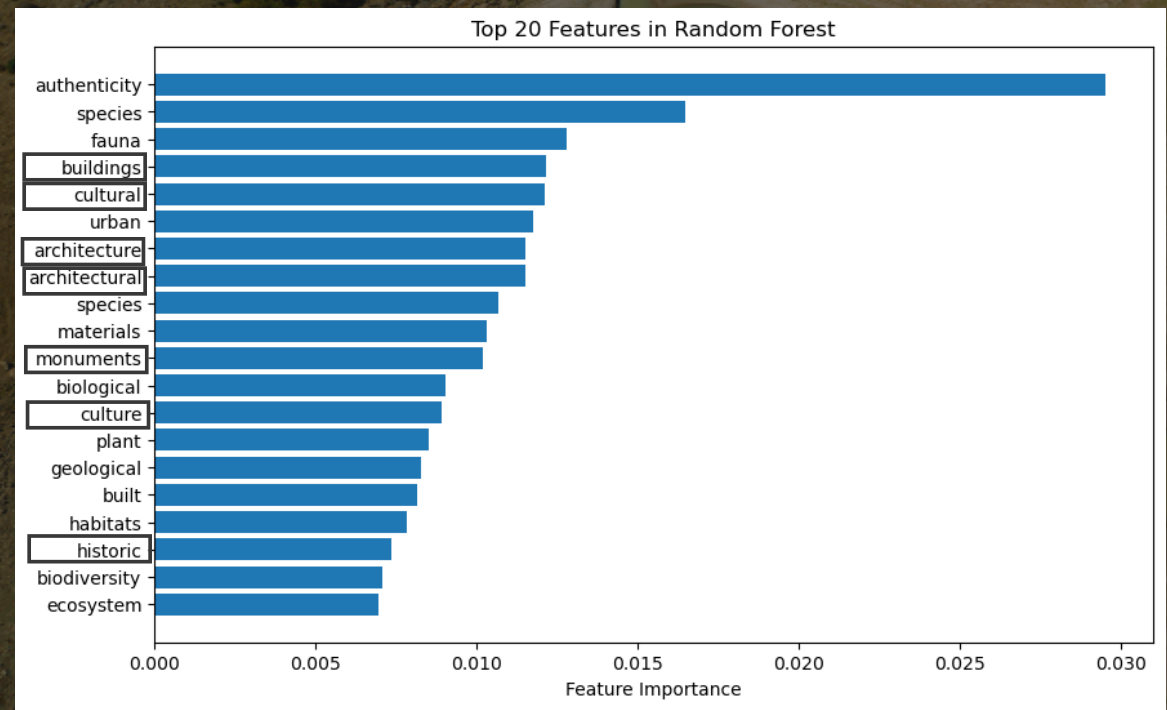
The model achieved a macro-F1 score of 0.99, indicating strong overall classification performance and suggesting a high degree of alignment between textual features and target labels.

Random Forest



ROC score of 99.98% shows that the model assigns a higher probability to the positive case, which means the model almost perfectly separates the two classes.

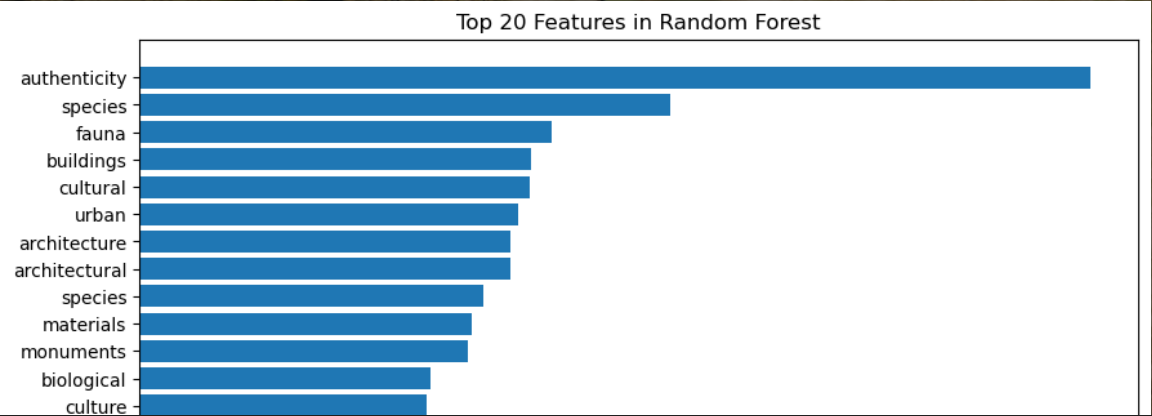
The high predictive performance across different evaluation metrics likely reflects a strong association between textual features and the target categories, which is justifiable since the description and justification texts should be a comprehensive introduction of a heritage site, highlighting its cultural/natural significance.



Random Forest

Random Forest's *feature_importances_* doesn't separate by class (Cultural or Natural) — it only tells the overall importance across all classes.

Certain words associated with cultural heritage—such as architecture, monuments—were strong predictors of the Cultural category, suggesting cultural indicators are more distinctive in the text.

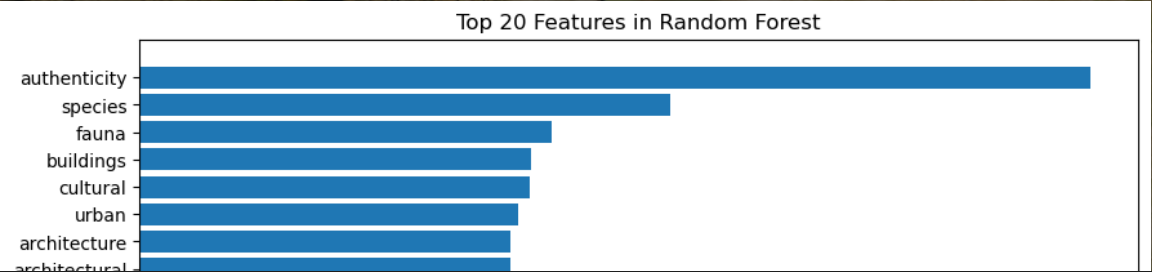


Random Forest

```
errors = df_test[df_test['label'] != df_test['pred_label']]
errors.head()[['Name EN', 'description', 'justification', 'label', 'pred_label']] # errors are all from Cultural (1) predicted as Natural (0)
```

Python

	Name EN	description	justification	label	pred_label
333	Rock-Art Sites of Tadrart Acacus	On the borders of Tassili NAjjer in Algeria al...		1	0
866	Mir Castle Complex	The construction of this castle began at the e...		1	0
331	Archaeological Site of Cyrene	A colony of the Greeks of Thera Cyrene was one...		1	0



Random Forest

```
df.loc[331]['description']  
# we can see some potential natural terminology like "earthquake".
```

Python

anized and remained a great capital until the earthquake of 365 A thousand years of history is written into its ruins which have been famous sinc

asymmetry in language signals

```
df.loc[333]['description']  
# we can see some potential natural terminology like "rocky".
```

Python

World Heritage site this rocky massif has thousands of cave paintings in very different styles dating from 12000 BC to AD 100 They reflect mark



Logistic Regression

Experiment with 3 hyperparameters:

- regularization strength C
- penalty type ($L1$ vs $L2$)
- solver algorithms (*liblinear* vs *saga*)

Logistic Regression

```
from sklearn.model_selection import GridSearchCV

rf = LogisticRegression(random_state=42)
hy_parameters = {'penalty': ('l1', 'l2'),
                 'C': [0.1, 1, 10],
                 'solver': ['liblinear', 'saga']} # 2*3*1 = 6 combinations

grid_cv = GridSearchCV(estimator = rf, param_grid = hy_parameters, cv=5)

grid_cv.fit(X_train_res, y_train_res)
```

```
from sklearn.model_selection import GridSearchCV

rf = LogisticRegression(random_state=42, class_weight='balanced')
hy_parameters = {'penalty': ('l1', 'l2'),
                 'C': [0.1, 1, 10],
                 'solver': ['liblinear', 'saga']} # 2*3*1 = 6 combinations

grid_cv = GridSearchCV(estimator = rf, param_grid = hy_parameters, cv=5)

grid_cv.fit(X_train, y_train)
```

The best configuration is:

- $L2$ regularization
- *saga* solver
- $C = 10$ (Undersampling), or
- $C = 1$ (using *class_weight='balanced'*)

Experiment with 3 hyperparameters:

- regularization strength C
- penalty type ($L1$ vs $L2$)
- solver algorithms (*liblinear* vs *saga*)

Logistic Regression

```
from sklearn.model_selection import GridSearchCV
```

```
rf = LogisticRegression(random_state=42)
```

```
hy_pa
```

		precision	recall	f1-score	support
grid_	Natural	1.00	1.00	1.00	82
	Cultural	1.00	1.00	1.00	276
grid_	accuracy			1.00	358

```
from skle
```

```
macro avg 1.00 1.00 1.00 358
```

```
rf = Logi
```

```
weighted avg 1.00 1.00 1.00 358
```

```
hy_param
```

```
'C':[0.1, 1, 10],
'solver': ['liblinear', 'saga']} # 2*3*1 = 6 combinations
```

```
grid_cv = GridSearchCV(estimator = rf, param_grid = hy_parameters, cv=5)
```

```
grid_cv.fit(X_train, y_train)
```

Python

The best configuration is:

- $L2$ regularization
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An aerial photograph of a two-lane asphalt road that curves through a dry, hilly landscape. The hills are covered in sparse, low-lying vegetation and small shrubs. The sky is overcast and grey. The overall tone of the image is muted and atmospheric.

04

Discussion

Model Comparison

Logistic Regression outperforms **Random Forest** in this classification task, likely due to the high-dimensional and sparse nature of TF-IDF features. **Logistic Regression** works very well in this kind of high-dimensional setting because it learns one overall decision boundary using all features together. In contrast, tree-based models like **Random Forest** are often better suited to structured tabular data rather than sparse text vectors. **Random Forest** makes decisions by splitting on individual features step by step. Since text data has thousands of weak features instead of a few strong ones, tree-based models may struggle to find clean splits. This can make **Random Forest** slightly less stable or less accurate compared to **Logistic Regression**.

Reflections

Based on peer feedback and previous project check-ins, I made several improvements to the code and analysis:

- Refined the **research question** to make it more realistic, given the dataset size and complexity.
- Addressed class imbalance using **undersampling** on the training data.
- Added **cross-validation and grid search CV** to ensure that model parameters were tuned systematically rather than chosen arbitrarily.
- Included **feature importance analysis and error analysis**, which help interpret the model's predictions and identify potential dataset biases.

These improvements strengthen both the technical validity of the models and the interpretability of the results.

Moving Forward

- *Incorporating more multimodal features, such as site regions and site images, to explore other relevant questions*
- *Applying advanced and unsupervised models like BERTopic to focus on potential discourses embedded in institutional datasets*
- *Exploring clustering methods to identify patterns in OUV Criteria combinations*



Thanks!